

KUALANAMU MEDAN, Indonesia

WMO: 960350

Lat: 3.640N

Lon: 98.879E

Elev: 25

StdP: 101.03

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.6	22.9	20.2	14.9	29.1	20.9	15.6	28.0	6.4	28.9	5.9	29.1	0.7	240	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.0	34.2	26.1	33.7	26.1	33.1	26.1	27.9	31.3	27.5	31.1	27.1	30.8	3.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	29.6	26.6	22.2	29.3	26.2	21.6	29.1	89.5	31.4	87.5	31.2	86.0	31.1	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	5.0	21.1	35.9	0.8	1.0	20.6	36.7	20.1	37.3	19.6	37.8	19.0	38.6		
			20.2	28.7	0.8	0.8	19.6	29.2	19.2	29.7	18.7	30.1	18.2	30.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.2	27.6	28.1	28.4	28.5	28.5	28.2	27.9	27.6	27.4	27.3	27.1
	DBStd	0.98	0.83	0.84	0.81	0.87	0.89	0.90	0.88	0.90	0.83	0.83	0.73	0.84
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6502	533	492	562	552	574	553	563	555	528	541	520	529
	CDD18.3	3460	274	259	303	302	316	303	304	297	278	282	270	271
	CDH23.3	34857	2575	2594	3223	3281	3441	3309	3194	2986	2626	2630	2529	2471
(13)	CDH26.7	13038	821	937	1299	1338	1410	1387	1312	1171	962	878	794	728
(14)	Wind	WSAvg	1.7	1.7	1.7	1.6	1.6	1.7	1.7	1.7	1.7	1.6	1.6	1.7
(15) Precipitation	PrecAvg	2549	158	99	151	148	221	171	175	227	317	341	270	271
	PrecMax	3747	369	289	385	344	451	372	313	447	571	634	509	512
	PrecMin	1636	72	40	34	24	79	78	59	103	171	188	102	110
	PrecStd	523	70	58	96	79	96	69	69	84	95	95	89	116
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	33.2	34.1	34.5	35.0	35.0	34.8	34.4	33.6	33.2	32.8	32.2
		MCWB	25.8	26.0	26.2	26.7	26.3	25.4	25.4	25.7	26.0	25.8	25.8	25.7
	2%	DB	31.9	32.5	33.2	33.5	33.9	33.8	33.7	33.2	32.7	32.1	31.9	31.4
		MCWB	25.6	25.8	26.0	26.5	26.6	26.1	25.7	25.9	25.9	25.8	25.8	25.7
	5%	DB	31.1	31.9	32.6	32.9	33.0	33.0	32.9	32.4	31.9	31.3	31.1	30.9
		MCWB	25.6	25.6	26.0	26.4	26.5	26.2	25.9	25.9	25.8	25.8	25.8	25.7
	10%	DB	30.4	31.1	31.9	32.0	32.1	32.1	32.0	31.7	31.1	30.8	30.3	30.1
		MCWB	25.5	25.6	26.0	26.2	26.4	26.2	25.8	25.9	25.9	25.8	25.8	25.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.2	27.7	28.0	28.2	28.2	28.0	27.9	27.6	27.6	27.4	27.2
		MCDB	30.6	31.2	31.6	31.9	31.9	31.7	31.0	31.4	30.6	30.5	29.9	30.0
	2%	WB	26.6	26.7	27.1	27.5	27.7	27.6	27.2	27.1	26.9	26.7	26.7	26.7
		MCDB	29.9	30.5	31.1	31.4	31.4	31.2	30.7	30.8	30.5	30.0	29.8	29.5
	5%	WB	26.2	26.4	26.7	27.0	27.2	27.1	26.7	26.6	26.5	26.3	26.4	26.2
		MCDB	29.4	30.0	30.7	31.1	31.0	30.9	30.4	30.4	30.1	29.6	29.4	29.1
	10%	WB	25.9	26.0	26.4	26.6	26.8	26.6	26.4	26.2	26.1	26.0	26.1	25.9
		MCDB	29.1	29.6	30.2	30.6	30.5	30.4	30.1	30.0	29.7	29.3	29.1	28.8
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.7	8.1	8.0	8.0	8.3	8.4	8.1	7.7	7.1	6.7	6.5
		MCDBR	7.9	8.7	9.0	8.9	9.0	9.2	9.5	9.2	8.7	8.1	7.7	7.6
		MCWBR	3.2	3.4	3.4	3.4	3.4	3.5	3.6	3.6	3.5	3.2	3.1	3.1
	5% WB	MCDBR	7.2	7.8	8.2	8.2	8.1	8.3	8.5	8.4	7.9	7.2	6.9	6.7
		MCWBR	3.2	3.3	3.5	3.4	3.5	3.6	3.7	3.6	3.6	3.2	3.1	3.1
(40) Clear-Sky Solar Irradiance	taub		0.477	0.524	0.504	0.473	0.466	0.479	0.469	0.489	0.488	0.498	0.453	0.453
	taud		2.255	2.102	2.175	2.272	2.309	2.274	2.303	2.231	2.235	2.203	2.342	2.340
	Ebn at Noon		850	817	832	842	827	803	816	817	832	829	866	865
	Edh at Noon		141	168	156	139	130	132	130	143	145	150	129	128
(44) All-Sky Solar Radiation	RadAvg		4.42	4.88	5.10	4.85	4.77	4.86	4.74	4.74	4.55	4.35	4.12	4.05
	RadStd		0.26	0.25	0.27	0.21	0.27	0.28	0.24	0.33	0.36	0.49	0.19	0.28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.19	+0.11	N/A	N/A	+0.40	+0.56	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.19	+0.11	N/A	N/A	+0.40	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page